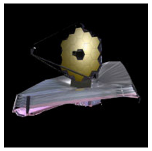

5409 - A Fundamental Study of Star and Planet Formation: Spectroscopic Confirmation of Free-floating Jupiter Mass Objects in NGC 2024

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	Jan21Plan_3step	NIRSpec MultiObject Spectroscopy	(12) source_list_v2

ABSTRACT

Theoretical star formation models have estimated the fundamental limit of turbulent fragmentation of molecular cloud cores to be ~ 1-10Mjup, overlapping masses that can result from planet formation within the disk of a star. At the same time, observational studies identify a decreasing initial mass function (IMF) approaching the limit of formation, yet are incomplete below ~ 20 Mjup. Recent JWST/NIRCam observations in NGC 2024, a moderate density embedded cluster with an age < 1 Myr, have identified dozens of candidate planetary mass objects with estimated masses down to 0.7 Mj based on their photometric color. We propose to obtain 1-5 micron spectra of 75 candidate planetary mass objects and brown dwarfs with estimated masses < 40 Mjup (45 of which are < 10 Mjup) in NGC 2024 with the NIRSpec/MSA in prism mode. With these spectra, we will use both atmospheric models and previously observed planetary mass object spectra in order to estimate masses and determine membership of our sample. Additionally, we will utilize the spectral retrieval technique to estimate elemental abundances and search for signatures of formation within a circumstellar disk, e.g. reduced C/O ratio due to formation beyond the H2O ice line. Our program requires only 10.7 total hours in order to: 1) probe the low mass limit of turbulent fragmentation (i.e. brown dwarf formation), 2) define the shape of the IMF down to Jupiter mass scales, and 3) identify Jupiter-mass ejected planets from their enhanced elemental abundances relative to other cluster members.

OBSERVING DESCRIPTION

We propose to obtain NIRSpec/MSA prism (R~100) spectra of at least 75 candidate planetary mass objects and brown dwarfs in NGC 2024. NIRCam imaging has already been performed in this star-forming region, resulting in precise astrometric measurements of the locations of these sources, and therefore we require only NIRSpec/MSA observations without additional NIRCam imaging. In this program, we will observe 75 sources over 1-5 microns using six MSA configurations in four visits. We require a total exposure time of 2801s using the three shutter slitlet for each source and reading out with the NRSIRS2RAPID pattern with 15 groups/integration and 4 integrations/exposure and 3 dithers. This achieves higher sensitivity than default readout patterns, required for our observations of faint sources. These observations will achieve S/N > 5 across 1-5 microns, differentiate elemental abundance features at the 5 sigma level, and achive S/N > 30 in dominant spectral features of high and low surface gravity for a typical 1 Mj source. In addition, we request NIRCam parallel imaging in many medium and broadband filters that will observe a separate part of the region than previously studied in order to identify other candidate planetary mass objects. Here, we will use the shallow2 readout pattern with 3 groups/integration, 6 integrations/exposure, and 3 dithers to identify a 1 Mj object through Av=30 mag with S/N=5 in the F430M filter. These parallel observations will also provide for future studies of environmental effects on star formation.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	source_list_v2	RA: 05 41 41.7719 (85.4240496d) Dec: -01 53 52.32 (-1.89787d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 5409, Observation 1: Jan21Plan_3step

Wed Jan 22 20:00:17 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Coordinated Parallel Template(s): NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Data Excess over middle threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Data Excess over lower threshold

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:3) Warning (Form): Data Excess over lower threshold

(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:3) Warning (Form): The recommended value is 8 Reference Stars for this template.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(12)	source_list_v2	RA: 05 41 41.7719 (85.4240496d) Dec: -01 53 52.32 (-1.89787d) Equinox: J2000		

Comments:
Description=[]

Acquisition

NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
3	Filter: F140X; Readout: NRSRAPID; 7 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

NIRSpec MultiObject Spectroscopy

TA Method: MSATA

HFF Readout Mode: false

Obtain Confirmation Images: After Target ACQ and New MSA Config

Science Aperture: MSA Center

Primary Candidate List: source_list_v2 (689 sources)

Filler Candidate List: null

Spectral Overlap Map: jwst-nirspec-prism

Spectral Overlap Threshold: 1.5

NIRCam Imaging

Module: ALL

Subarray: FULL

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	148	85.434121	-1.922381	20.7	1	238	85.449868	-1.899623	22.029
1	186	85.419342	-1.896791	21.251	1	239	85.456502	-1.906068	22.038
1	209	85.416234	-1.894460	21.565	1	240	85.455480	-1.920743	22.055
1	214	85.433001	-1.891487	21.618	1	249	85.409163	-1.922444	22.171

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
2	179	85.455422	-1.879020	21.103	2	208	85.438883	-1.917462	21.562
2	194	85.456805	-1.918268	21.402	2	215	85.423497	-1.903465	21.633
2	200	85.456559	-1.918430	21.463	2	219	85.469110	-1.887635	21.665
2	201	85.422456	-1.897903	21.464	2	223	85.467386	-1.894402	21.764

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
3	158	85.441599	-1.917287	20.898	3	209	85.416234	-1.894460	21.565
3	186	85.419342	-1.896791	21.251	3	251	85.446007	-1.919863	22.184
3	198	85.401363	-1.896418	21.445	3	320	85.402643	-1.930983	22.91
3	208	85.438883	-1.917462	21.562					

Dithers

#	Dither Type
1	NONE

Confirmation

NIRSpec MultiObject Spectroscopy	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time
1	c1	NRSIRS2RAPID	5	1	1	87.533
2	c3	NRSIRS2RAPID	5	1	1	87.533
3	c4	NRSIRS2RAPID	5	1	1	87.533
4	c2	NRSIRS2RAPID	5	1	1	87.533

Spectral Elements

NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	85.434636458333 34 Degrees -1.9142258333333 189 Degrees	215.02448708739 17			3	15	5252.0
2	1 (PRISM/CLEAR)	c2	3 Shutter Slitlet	85.41981325 Degrees -1.9150366666666 514 Degrees	215.02497242479 44			3	15	5252.0
3	1 (PRISM/CLEAR)	c3	3 Shutter Slitlet	85.442094541666 66 Degrees -1.8919741666666 71 Degrees	215.02419994310 432			3	15	5252.0
4	1 (PRISM/CLEAR)	c4	3 Shutter Slitlet	85.414823541666 68 Degrees -1.9207000000000 107 Degrees	215.02514649753 326			3	15	5252.0

Spectral Elements

NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F182M	F430M	SHALLOW2	8	4	12	3	4863.757	
2	F182M	F430M	SHALLOW2	8	4	12	3	4863.757	
3	F182M	F430M	SHALLOW2	8	4	12	3	4863.757	
4	F182M	F430M	SHALLOW2	8	4	12	3	4863.757	

Special Requirements

Group Visits within 53.0 Days
Visits Same PA
No Parallel Attachments
MSA Scheduled Aperture PA 215.0248 to 215.0248 Degrees (V3 76.45025 to 76.45025)